

Smell and taste

Our senses of smell and taste both involve collecting chemicals and analysing them. The nose collects chemicals carried in the air. Inside the nose, scent chemicals dissolve in the mucus lining of the nasal cavity (which also helps clean the air). The chemicals are detected by hairlike nerve endings that send signals to the brain. The tongue detects similar chemicals in food.

▷ Taste bud

Taste buds are located on the tongue, gums, and throat. They have nerve endings covered in proteins that can detect specific chemicals associated with certain foods, such as sweet sugar or sour acids.

the nerve endings send signals from the taste receptor cells to the brain

olfactory bulb takes signals from the nose to the brain

nasal conchae force inhaled air to flow steadily

the nasal cavity is shaped to increase the size of the odor-sensitive layer

tastebuds on tongue can detect five distinct tastes: sweet, sour, bitter, salty, and umami (savory)

nerves under the tongue's surface take taste signals to the brain

openings in the tongue called taste pores allow food dissolved in saliva to reach the taste receptors

each taste receptor cell picks up certain chemicals such as sugars (sweet) and acids (sour)

▷ Skin

Mechanoreceptors pick up pressure, touch, stretching, vibration, and sharp pains. Heat and cold are picked up by thermoreceptors.

heat receptor

cold receptor

nerves send messages to the nervous system

sebaceous gland secretes oily sebum to lubricate skin and hair

sweat gland secretes sweat to cool the skin

touch receptor

pain receptors

sweat pore

pressure receptor

hair shaft

Touch

The sense of touch relies on several types of receptors, mainly located in the skin, but also found in muscles, joints, and internal organs. There are about 50 touch receptors for every square inch of skin, although more sensitive body parts, such as the fingertips and tongue, have more, while the back has fewer.

REAL WORLD

Braille

Fingertips (touch), and not eyes (vision), are used to read Braille. Letters are represented by patterns of between one and five small bumps, or dots, arranged in a grid. Skilled Braille readers can read about 200 words per minute.

